

Overview of the initiative

Please provide objective.

T-Mobile is building an enterprise-wide **DEX (Developer Experience Platform)** to centralize, standardize, and measure developer tools, services, and workflows across the organization. The initiative aims to create a unified ecosystem that improves developer productivity, provides comprehensive metrics and observability, and enables data-driven decision-making for tool investments and platform improvements.

Primary Goals:

Unified Developer Experience - Single pane of glass for all developer tools and services

Metrics & Observability - Comprehensive measurement of developer productivity and tool adoption

Tool Standardization - Consistent patterns across all development tools

Cost Optimization - Track ROI and eliminate redundant tools

Security & Compliance - Centralized governance and audit capabilities

Developer Productivity - Reduce friction and accelerate time-to-production

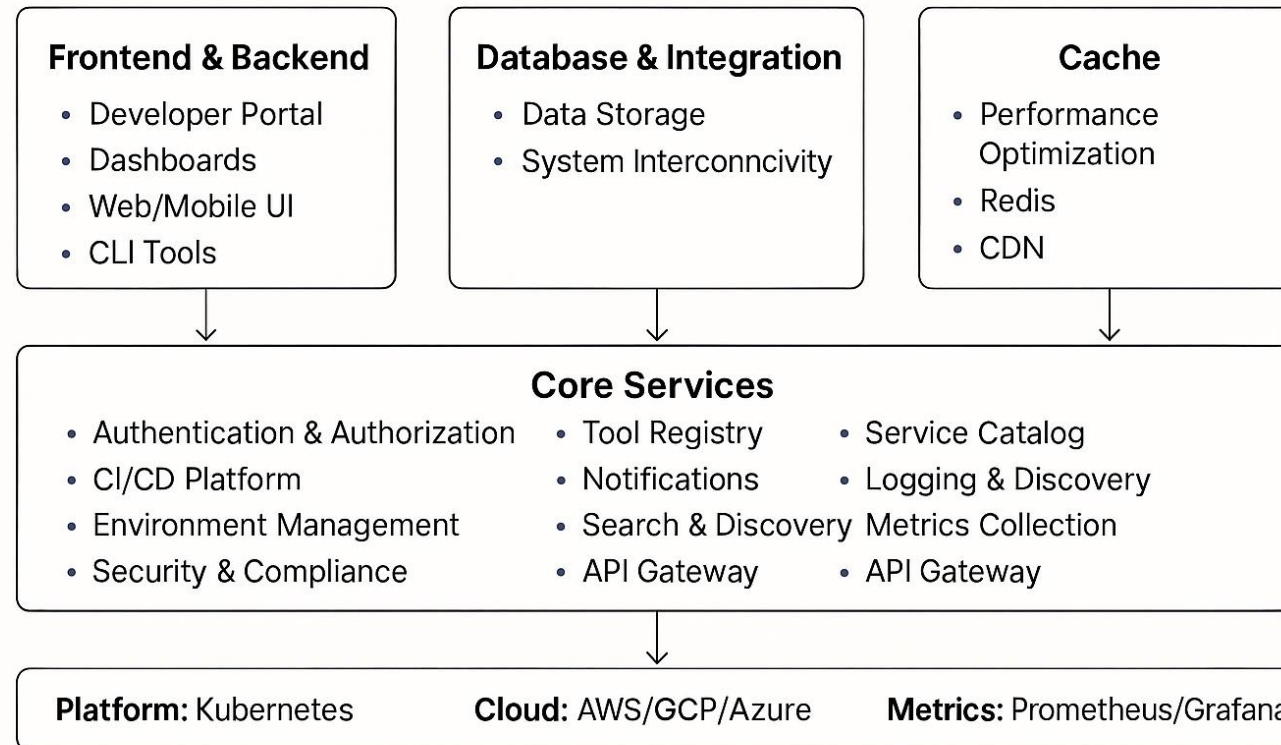
Example Response

This initiative is to support Home Internet customers to improve authentication service for at home routers/devices. When an end-user requires TMO's internet service, wholesale resellers like Sify install the required device at the user's home or office. These devices tend to authenticate to Sify using their MAC address to get their public IPv4 and IPv6 addresses from Ciena which is controlled by TMO public IP team. We need to deploy authentication service in TMO cloud as this being a critical system and left to be accessible from any Ciena data center in a secured way.

Provide your call flow diagram

Delete the previous slide once you have provided your diagram.

Dex – Architecture



Cloud Services

Please list all services you intend on deploying.

EKS, S3, RDS, ALB , VPC end point, Route53

Example Response

AWS Site 2 Site VPN • EC2 Instances • S3 Storage • Private Link Service • VPC Endpoint Service

HA / DR

Please provide High availability requirement (Active – Active/DR, Regions etc).

EKS cluster deployed across multiple AZs (3 availability zones minimum).

Example Response

The solution will be deployed in two regions to support High availability and redundancy. AWS West 2 region will be primary location with AWS East 1 as DR.

Integrations

Please provide internal and external integration requirements.

The IDP platform will integrate with T-Mobile's internal systems including GitHub Enterprise for source code management, Jira for project tracking, and Jenkins/GitHub Actions for CI/CD pipelines. The platform will connect to internal Artifactory for artifact storage and deploy to multi-cloud environments (AWS, , Azure) via VPC peering. Integration with ServiceNow for incident management and notifications. All internal traffic routed through T-Mobile private network. Splunk APM Prometheus and Grafana for metrics aggregation. Secret management via HashiCorp Vault.

Example Response

The customer traffic come from internet to partner data center and then to T-Mobile cloud via site 2 site VPN. This solution need to connect to Radius server in T-Mobile P8 VPC (via Private link service). Couple of EC2 instances to deploy integration services. S3 storage will be stored to logs

Roles / Services Accounts

Please provide if any special identities required ex: AWS service account, access key etc.

IDP platform deployment uses Infrastructure as Code (Terraform/Pulumi). Need AWS service account with IAM role for EKS cluster admin, RDS database creation, and S3 bucket management. If OIDC is supported, require role with EKS, RDS, S3, VPC admin privileges to create and deploy platform services. gitlab service account for CI/CD pipeline integration. Service accounts for connecting to gitlab Enterprise, Jira APIs, and Artifactory with appropriate read/write permissions.

Example Response

We will be deploying Infra as Code so need AWS access key/secret to configure git pipeline. If OIDC is supported then we need role with Ec2, S3 admin privileges to to create and deploy services.

OS / Images

Please provide OS and images required.

T-mobile standard RHEL images

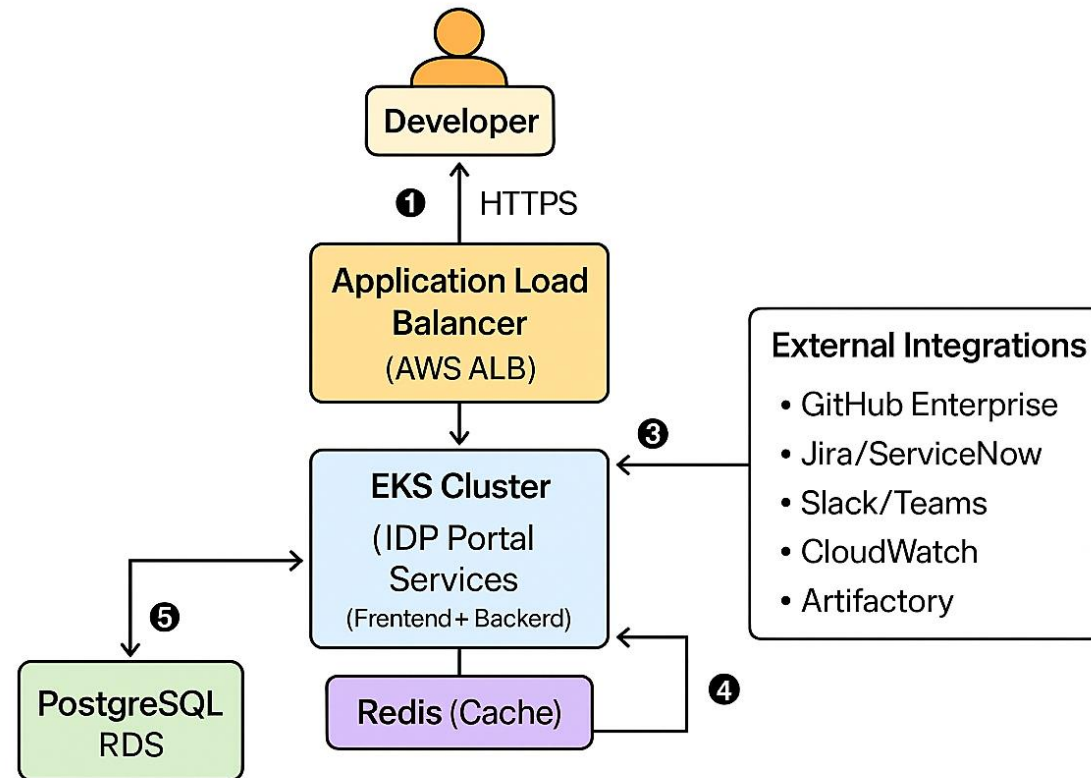
Example Response

We will leverage T-Mobile standard linux (RHEL) images to deploy VM's.

Provide your call flow diagram

Delete the previous slide once you have provided your diagram.

DEX Platform Call Flow



Data Types

Please list all data types (Ex: MSISDN, Name, Location, Address etc) application is using.

Example Response

Only Radius service will be used to authenticate with MAC address
CPNI – None • PCI – None • PII – None • Other – Device MAC address

Encryption

Please include how data is protected and encryptions if using.

Example Response

Though Radius protocol is not encrypted, we are routing radius traffic via Site-site VPN.



Overview

Please provide application functions to be deployed on Container Platform.

IDP platform provides centralized developer portal for tool discovery, service catalog management, CI/CD pipeline orchestration, metrics collection and observability dashboards. Core services include authentication/authorization, tool registry, environment management, and integration hub for gitlab, Jira, and monitoring systems.

Example Response

Please state if you have any dependencies in applications or services deployed on public Cloud.

Cluster / Container Requirements

Please provide capacity requirements.

Need 3 environments conductors dev, stg, prod. **Right now we need Dev konduktor ASAP**

Per Environment Summary:

- Total Pods: ~2 5-30 pods (frontend, backend APIs, services, cache, monitoring)
- CPU: ~70-80 vCPU total
- Memory: ~150-170GB total
- Storage: 500GB PVC for logs and persistent data.

Example Response

No. of Pods/CPU/Memory • Dedicated nodes (if required) etc. • Environments / Stacks • No. of Environments (Dev, Stg, Perf, Prod etc.)
Scaling requirements



Other Requirements

Please provide details about additional cluster requirement if any.

Do you need Cluster level access?

Yes , Initially required for setup,, RBAC configuration, and installing platform-wide components (Istio, cert-manager, PV , LoadBalancer, ingress controller etc) . Post-deployment, namespace-level admin access sufficient for ongoing operations and application management.

Do you need any custom CRD, Webhooks etc installs?

Admission webhooks for pod security policies, Cert-manager validation webhooks, Istio sidecar injection webhooks, Istio CRDs for service mesh configuration, flagger operator, Vault Secrets Operator, SecretProviderClass CRD. OTEL enabled along with Prometheus.

If any additional objects or CRD required will request to conductor team.

Do you need any additional namespace access, Cluster roles etc

Yes, need possible roles for coduktors env with admin.



Application Requirements

Please provide Kubernetes API access requirements (if required).

Any daemonset requirements?

Logs and metrics

Non-HTTPS traffic?

External traffic is always with HTTPS.

GPUs?

No - IDP platform does not require GPU resources